Chaoqi Zhang

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Research Interests

My research lies at the intersection of **time series forecasting** and **machine learning for healthcare**. I aim to build forecasting models that are accurate, efficient, and reliable, and to apply them to real-world clinical data, with the broader goal of developing trustworthy **foundation models for healthcare**.

Technical Skills

- **Programming**: Python, C/C++, Java, SQL, Shell
- **ML & Data Tools**: PyTorch, JAX, Hugging Face Transformers, scikit-learn, NumPy, pandas
- **ML Methods**: Time series modeling, LLM and multimodal evaluation, retrieval-augmented generation (RAG), adversarial robustness
- **Systems & Development**: Linux, Git, Docker, Apache Flink, Apache Kafka, PostgreSQL, Flask, React, KLEE

Education

- **Indiana University Bloomington** — Bloomington, IN
  Ph.D. in Computer Science — *Aug. 2022 – Dec. 2027 (Expected)*
  Advisor: Prof. Haixu Tang.
- **The Hong Kong University of Science and Technology** — Hong Kong
  M.Sc. in Information Technology — *Sep. 2019 – Nov. 2020*
- **University of Electronic Science and Technology of China** — Chengdu, China
  B.Eng. in Communications Engineering — *Sep. 2011 – Jul. 2015*

Research and Professional Experience

- **Indiana University Bloomington** — Bloomington, IN
  Research Assistant — *Aug. 2022 – Present*
  - Develop efficient and reliable forecasting models for temporal and physiological data, combining theoretical analysis, model design, and empirical evaluation.
  - Build LLM benchmarks and evaluation pipelines for scientific applications; develop tools for robotic software verification, adversarial robustness, and software privacy.
- **The Chinese University of Hong Kong** — Hong Kong
  Research Assistant — *Feb. 2022 – Aug. 2022*
  - Built an automated bug-detection tool using KLEE symbolic execution to uncover vulnerabilities in network protocol software.
- **The Hong Kong University of Science and Technology** — Hong Kong
  Research Assistant — *Jul. 2020 – Jan. 2022*
  - Built a real-time continuous-query system with Apache Flink and Kafka, a Flask backend, and React visualizations (**VLDB 2021 demo**).
- **TP-Link Technologies Co., Ltd.** — Shenzhen, China
  Software Engineer, SMB Department — *Aug. 2015 – May 2017*
  - Led full-stack development of an enterprise Radio Calibration & Test System, tripling production throughput across 13 wireless hardware products while coordinating four engineers.

Publications

[*] Equal contribution (co-first authors).

[1] **When2What: A Timing-Conditioned Framework for Long-Horizon Forecasting of Continuous Cardiovascular Waveforms**
Chaoqi Zhang, Shujie Yan, Zhao Zhang, Yanhua Sun, Yu Wang, Li Li, Lingzhong Meng, Xiaoyi Chen, Haixu Tang.
arXiv preprint arXiv:2611.08993, 2026. [arXiv] [PDF] [Code]

[2] **From HL to $H + L - 1$ Parameters: A Hankel–Toeplitz Forecaster for Long-Term Time Series Forecasting**
Chaoqi Zhang, Yu Wang, Haixu Tang.
arXiv preprint arXiv:2609.33984, 2026. [arXiv] [PDF]

[3] **LLMs as Lab Engineers: A Benchmark for Analytical Method Lifecycle Management**
Xiaoyi Chen[*], Mahsa Monshizadeh[*], **Chaoqi Zhang**, Jianjun Lang, Yang Wu, Genevieve Mortensen, Xiaozhong Liu, Haixu Tang.
Findings of the Association for Computational Linguistics: ACL 2026, pp. 40533–40553, 2026. [Paper] [PDF]

[4] **RVSPEC: Cyber-Physical Interplay Graphs for Formal Specification of Robotic Vehicle Control Software**
Chaoqi Zhang, Minhyun Cho, Inseok Hwang, Hyungsub Kim.
IEEE International Conference on Robotics and Automation (**ICRA 2026**).

[5] **Avara: A Uniform Evaluation System for Perceptibility Analysis Against Adversarial Object Evasion Attacks**
Xinyao Ma[*], **Chaoqi Zhang**[*], Huadi Zhu[*], Jean Camp, Ming Li, Xiaojing Liao.
ACM SIGSAC Conference on Computer and Communications Security (**ACM CCS 2024**). [Paper]

[6] **Measuring Compliance Implications of Third-party Libraries' Privacy Label Disclosure Guidelines**
Yue Xiao, **Chaoqi Zhang**, Qin Yue, Fares Fahad S Alharbi, Luyi Xing, Xiaojing Liao.
ACM SIGSAC Conference on Computer and Communications Security (**ACM CCS 2024**). [Paper]

[7] **iHunter: Hunting Privacy Violations at Scale in the Software Supply Chain on iOS**
Dexin Liu, Yue Xiao, **Chaoqi Zhang**, Kaitao Xie, Xiaolong Bai, Shikun Zhang, Luyi Xing.
33rd USENIX Security Symposium (**USENIX Security 2024**). [PDF]

[8] **Cquirrel: Continuous Query Processing over Acyclic Relational Schemas**
Qichen Wang, **Chaoqi Zhang**, Danish Alsayed, Ke Yi, Bin Wu, Feifei Li, Chaoqun Zhan.
Proceedings of the VLDB Endowment, 14(12):2667–2670, 2021. **VLDB 2021, Demonstration**. [PDF] [Code]

Posters

[1] **A Multi-Agent Framework for Formal Specification of Robotic Vehicle Control Software**
Chaoqi Zhang, Hyungsub Kim.
3rd USENIX Symposium on Vehicle Security and Privacy (**VehicleSec 2025**), poster abstract. [Paper] [PDF]
Best Poster Award, Midwest Security Workshop 2025.

Academic Service

- Reviewer: *IEEE ICASSP* (2027), *IEEE S&P* (2023–2025), *RAID* (2025), *ACM CCS* (2023), *ACSAC* (2023).
- Artifact Evaluation Committee: *NDSS* (2026), *ACM CCS* (2025), *ACSAC* (2022, 2023).

Teaching

- **Assistant Instructor, Indiana University Bloomington** *2022–2026*
  Advanced Operating Systems; Computer Networks; Systems & Protocol Security & Information Assurance; Security for Networked Systems; Data Analysis and Mining; Big Data Analytics.

Awards and Honors

- **Best Poster Award**, *Midwest Security Workshop* *2025*
- National Encouragement Scholarship, *UESTC* *2012, 2013*